

Lecture 25

Random-Access Memory

Memory units · address and word · read and write · timing and wait states · SRAM and DRAM · inside a RAM

0	1	1	0
1	0	1	1
0	0	0	1
1	1	0	0

address 01 → word 1011

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 7, §7.1–7.2

Today's roadmap

1

§7.1

Memory units

Words, bits, address lines

2

§7.2

Read and write

Operations and timing

3

§7.2

SRAM and DRAM

Latch or capacitor

4

§7.2

Inside a RAM

Cells, decoder, array

By the end of class you can...

- Relate words, word length and address lines
- Describe the read and write operations
- Compute wait states from access time and clock period
- Compare SRAM and DRAM cells
- Draw the internal organization of a small RAM

Agenda for today's session

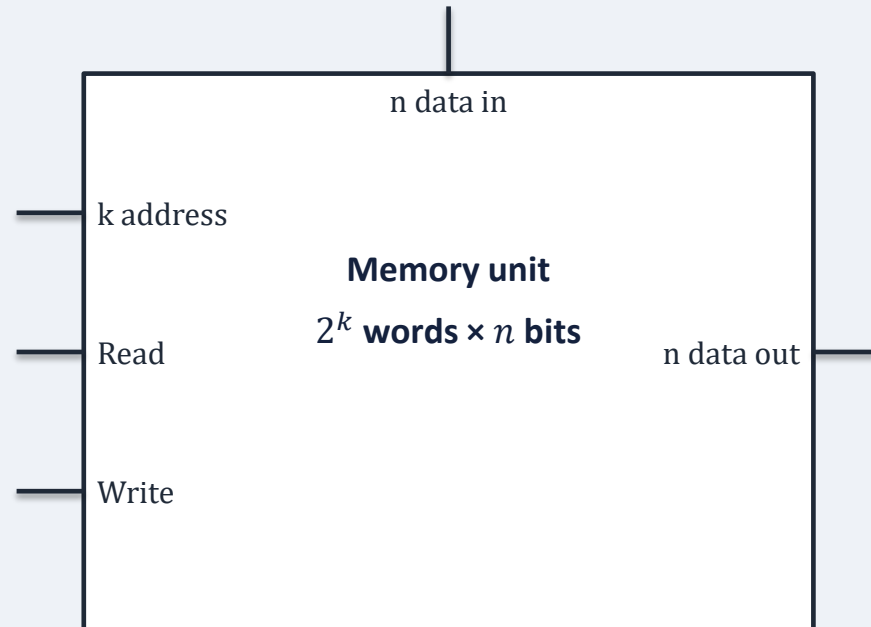
Intro	Units	R/W	Break	Cells	Array	Wrap	
0:00	0:05	0:25	0:50	1:00	1:15	1:40	1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:25	Memory units and addressing	Examples 1, 2
0:25 – 0:50	Read, write and timing	Example 3 · Your turn 1
0:50 – 1:00	Break	
1:00 – 1:15	SRAM and DRAM cells	—
1:15 – 1:40	Binary cell and RAM array	Example 4 · Your turn 2
1:40 – 1:50	Verilog; concept check; wrap-up	—

A memory unit

§7.1

Block diagram



The one relationship

k address lines select 2^k words

n data lines set the word length

Capacity = $2^k \times n$ bits

Units

K = 2^{10} , M = 2^{20} , G = 2^{30} in memory sizes.

RAM: read and write. ROM: read only (Lecture 27).

Example 1 — Sizing memories

§7.1

EXAMPLE

Fill in the address lines, data lines and total bits.

Memory	Address lines	Data lines	Total bits
4K × 8	12	8	32,768
64K × 32	16	32	2,097,152
1M × 1	20	1	1,048,576
256 × 16	8	16	4,096

- 1 Address lines k : $2^k = \text{number of words}$
- 2 $4K = 4096 = 2^{12} \rightarrow 12$; $4096 \times 8 = 32,768$
- 3 $64K = 2^{16} \rightarrow 16$; $65,536 \times 32 = 2,097,152$
- 4 $1M = 2^{20} \rightarrow 20$ (10 if multiplexed)
- 5 $256 = 2^8 \rightarrow 8$; $256 \times 16 = 4,096$
- 6 Data lines = the word width n
- 7 Words $\neq$ bits: read the question

Example 2 — Address and contents

§7.1

EXAMPLE

Word number 565 of a $1K \times 16$ memory holds 1210. Give the address and contents in binary.

10-bit address

512	256	128	64	32	16	8	4	2	1
1	0	0	0	1	1	0	1	0	1

16-bit contents

0	0	0	0	0	1	0	0	1	0	1	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- 1 $1K = 1024 = 2^{10}$ words $\rightarrow$ 10 address lines; 16 data lines
- 2 $565 - 512 = 53$, $53 - 32 = 21$, $21 - 16 = 5$, $5 - 4 = 1$, $1 - 1 = 0$
- 3 1s at 512, 32, 16, 4, 1 $\rightarrow$ address **10 0011 0101**
- 4 $1210 - 1024 = 186$, $186 - 128 = 58$, $58 - 32 = 26$
- 5 $26 - 16 = 10$, $10 - 8 = 2$, $2 - 2 = 0$
- 6 1s at 1024, 128, 32, 16, 8, 2 $\rightarrow$ **0000 0100 1011 1010**
- 7 Check: convert both back to decimal ✓

Read and write operations

§7.2

Write

1. Put the address on the address lines.
2. Put the data on the data-in lines.
3. Assert Write; hold address and data stable.

Read

1. Put the address on the address lines.
2. Assert Read (and chip select).
3. After the access time, data appears at the output.

CS	R/W'	Operation
0	X	none (outputs Hi-Z)
1	0	write selected word
1	1	read selected word

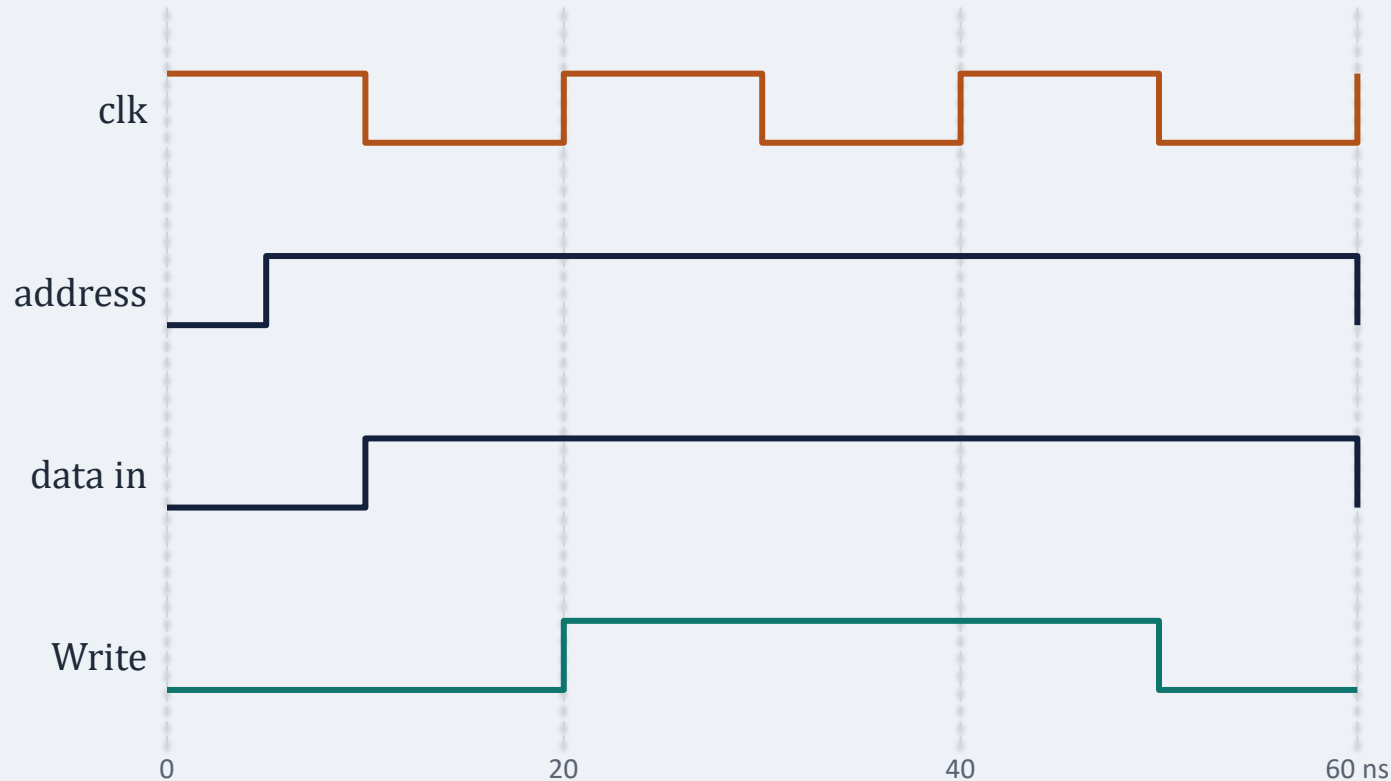
Many chips use one line

$R/W' = 1$ reads, 0 writes; chip select (CS) or enable turns the chip on. Outputs are three-state so many chips share one data bus.

Memory timing

§7.2

Write cycle (clock 50 MHz, cycle 50 ns)



Address and data become valid, Write pulses, and both stay stable until Write ends.

Definitions

Access time: valid address to valid data.

Cycle time: minimum time between successive operations.

Wait states

Clocks per access = $\lceil t_{cycle} / T_{clk} \rceil$

$50 / 20 = 2.5 \rightarrow 3$ periods (60 ns)

Example 3 — Clock periods per access

§7.2

EXAMPLE

A CPU runs at 150 MHz; the memory cycle time is 15 ns. How many clock periods does a read take?

- 1 Clock period: $T = 1/f = 1/150 \text{ MHz} = \mathbf{6.67 \text{ ns}}$
- 2 Periods needed: cycle time / $T = 15/6.67 = 2.25$
- 3 A read spans whole periods: round **up** $\rightarrow$ **3 periods**
- 4 $3 \times 6.67 = 20 \text{ ns} \geq 15 \text{ ns} \checkmark$
- 5 Two wait states beyond the first clock
- 6 Faster memory (cache) avoids them: the reason for SRAM caches

Your turn 1 — Size and speed

§7.2

YOUR TURN

≈ 10 min

- (a) An $8K \times 16$ memory: address lines, data lines, total bits?
- (b) A 100 MHz CPU reads a memory with a 35 ns cycle time. Clock periods per read?
- (c) A 200 MHz CPU and a 12 ns SRAM?
- (d) A $2K \times 4$ chip: how many bits, and how many address pins?

Always round clock periods up.

Your turn 1 — Solution

§7.2

SOLUTION

- 1 (a) $8K = 8192 = 2^{13} \rightarrow \mathbf{13}$ address lines
- 2 Word width 16 $\rightarrow \mathbf{16}$ data lines
- 3 $8192 \times 16 = \mathbf{131,072}$ bits
- 4 (b) $T = 1/100 \text{ MHz} = 10 \text{ ns}$
- 5 $35/10 = 3.5 \rightarrow$ round up: $\mathbf{4}$ periods
- 6 (c) $T = 1/200 \text{ MHz} = 5 \text{ ns}$; $12/5 = 2.4 \rightarrow \mathbf{3}$ periods
- 7 (d) $2048 \times 4 = \mathbf{8192}$ bits
- 8 $2048 = 2^{11} \rightarrow \mathbf{11}$ address pins
- 9 Check: capacity always equals $2^k \times n$

Break

10 minutes — we resume at the 1:00 mark

Next: SRAM, DRAM and the inside of a RAM

SRAM and DRAM

§7.2

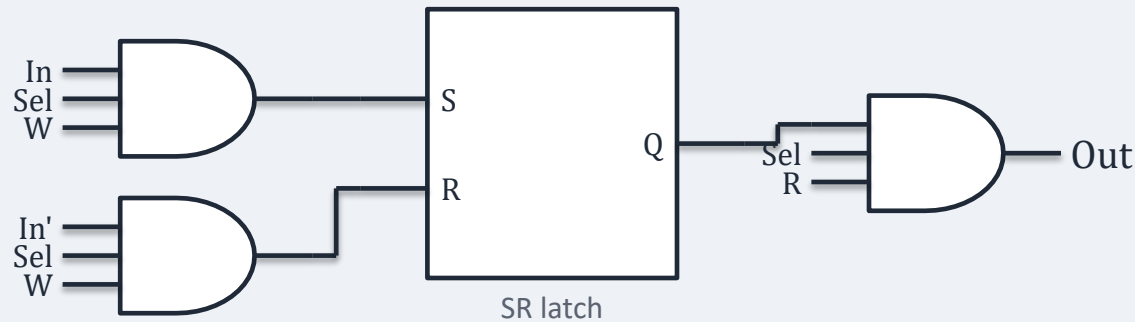
	Static RAM (SRAM)	Dynamic RAM (DRAM)
Cell	latch: about 6 transistors	1 transistor + 1 capacitor
Holds data	while powered	charge leaks: refresh
Speed	fast (a few ns)	slower (tens of ns)
Density, cost	low density, costly	high density, cheap
Used for	caches, FPGA block RAM	main memory

Refresh: every DRAM row is read and rewritten within about 64 ms. With 8192 rows that is one row every 7.81 μ s, stealing a little bandwidth.

The binary cell

§7.2

One bit: a latch plus select logic



W = write, R = read, Sel = this cell's word is selected.
In' comes from an inverter on In.

Behavior

Sel = 0: holds; output 0.

Sel = 1, write: latch set or reset to In.

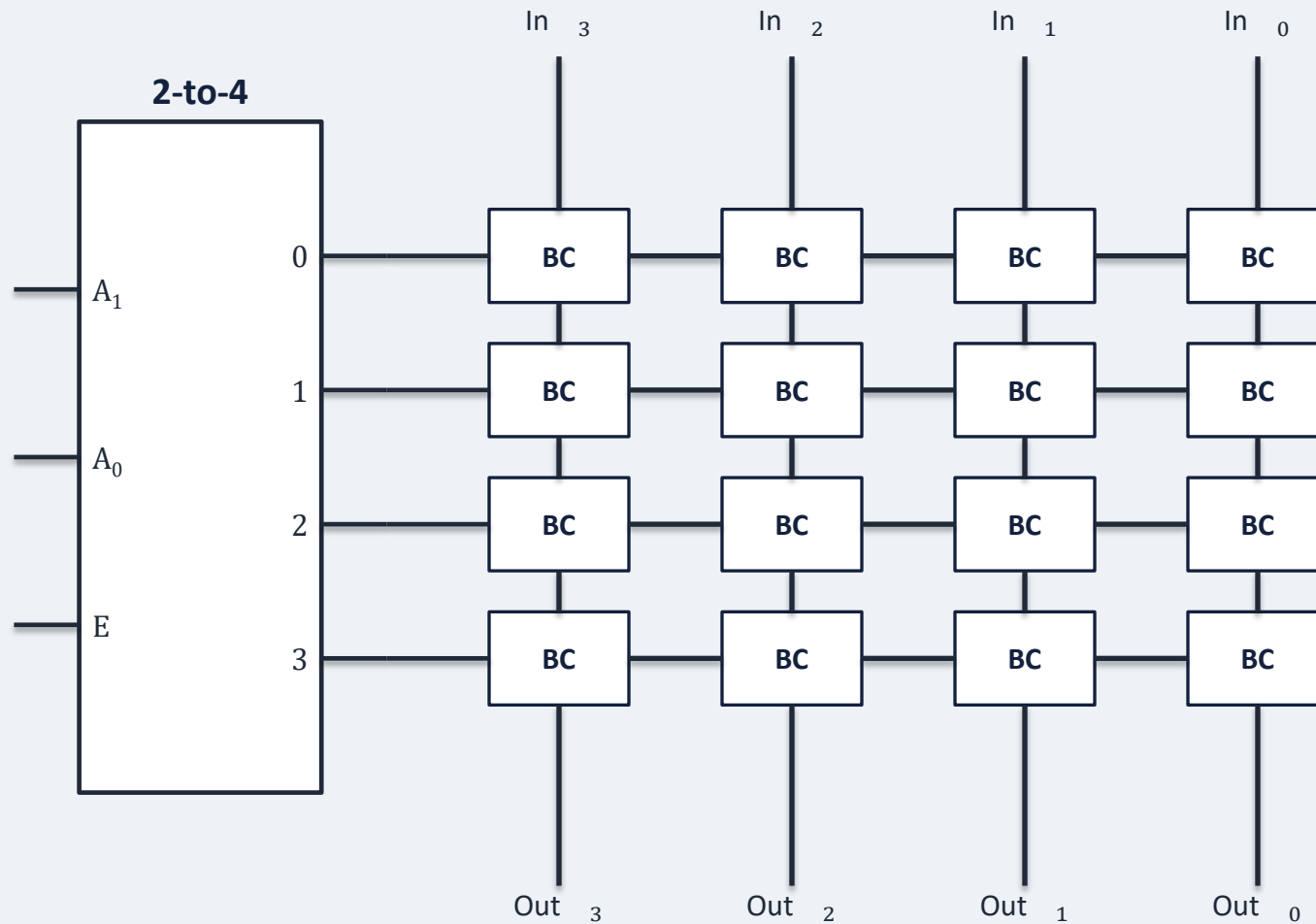
Sel = 1, read: Out = stored bit.

Symbol

A box BC with inputs Select, Read/Write, In and output Out.

Example 4 — Inside a 4×4 RAM

§7.2



How it works

The decoder selects one word (row).

Read/Write goes to every cell.

Each column's outputs are ORed onto one data line.

$2^k \times n$ RAM: a k -to- 2^k decoder and $2^k n$ cells.

Your turn 2 — Build a memory

§7.2

YOUR TURN

≈ 10 min

- (a) How many binary cells and what size decoder does a 16×8 RAM need?
- (b) In the 4×4 RAM, list the steps to write 1010 at address 2.
- (c) Why must the binary cell's select gate include the Read line?

Use the 4×4 structure as the template.

Your turn 2 — Solution

§7.2

SOLUTION

- 1 (a) $16 \text{ words} \times 8 \text{ bits} = \mathbf{128 \text{ cells}}$
- 2 $16 = 2^4 \text{ words} \rightarrow 4 \text{ address lines} \rightarrow \mathbf{a 4-to-16 \text{ decoder}}$
- 3 (b) Put the address $A_1A_0 = 10$ on the address lines
- 4 Put the data 1010 on In_3-In_0
- 5 Enable the decoder: only row 2 is selected
- 6 Assert Write: row 2 cells take 1, 0, 1, 0
- 7 Other rows are not selected: unchanged
- 8 (c) The cell's output gate needs Select AND Read
- 9 So a cell drives its column only when read
- 10 Unselected cells output 0, so the columns can be ORed

Memory in Verilog

§5.6

```
module ram16x8 (input clk, we,
                input  [3:0] addr,
                input  [7:0] din,
                output [7:0] dout);
    reg [7:0] mem [0:15];
    always @(posedge clk)
        if (we) mem[addr] <= din;
    assign dout = mem[addr];
endmodule
```

Notes

reg [7:0] mem [0:15] declares 16 words of 8 bits.

Writes happen on the clock edge when we = 1.

The read is combinational (asynchronous).

FPGA tools map this onto block RAM; a registered read is usually required for that.

Concept check

1 How many address lines does a $32\text{K} \times 8$ memory need?

2 What is the difference between access time and cycle time?

3 Why does DRAM need refresh and SRAM does not?

4 What does the decoder do inside a RAM?

Key takeaways

Words

2^k for k address lines

Capacity

$2^k \times n$ bits

Wait states

$\lceil t_{cycle}/T \rceil$ clocks

SRAM

latch, fast, no refresh

DRAM

capacitor, dense, refresh

Inside

decoder + binary cells

A RAM is a decoder driving an array of latches.

Homework #11 due at the start of Lecture 28

Next: memory decoding and error correction